Band-1 Hadamard order 14520 – Classical Paley-I (q=14519); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14520 Paley-I PASS; Paley-II FAIL/SKIP (q=7259=7*17*61 NOT prime; ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*3630; 3630===2 mod 4 NOT Hadamard-admissible; 3630=2*3*5*11^2); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-12

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-12_Band1_Hadamard_14508_PaleyII`; `2026-10-12_Band1_Hadamard_14504_PaleyI`; `2026-10-12_Band1_Hadamard_14480_PaleyI`; `2026-10-12_Band1_Hadamard_14476_PaleyII`; `2026-10-09_Band1_Hadamard_14460_PaleyII`; `2026-10-09_Band1_Hadamard_14448_PaleyI`; `2026-10-09_Band1_Hadamard_14432_PaleyI`; `2026-10-09_Band1_Hadamard_14428_PaleyII`; `2026-10-09_Band1_Hadamard_14424_PaleyI`; `2026-10-09_Band1_Hadamard_14420_PaleyI`; `2026-10-09_Band1_Hadamard_14412_PaleyI`; `2026-10-09_Band1_Hadamard_14408_PaleyI`; `2026-10-09_Band1_Hadamard_14388_PaleyI`; `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-10-09_Band1_Hadamard_14348_PaleyI`; `2026-10-09_Band1_Hadamard_14328_PaleyI`; `2026-10-09_Band1_Hadamard_14324_PaleyI`; `2026-10-09_Band1_Hadamard_14304_PaleyI`; `2026-10-09_Band1_Hadamard_14252_PaleyI`; `2026-10-09_Band1_Hadamard_14244_PaleyI`; `2026-10-09_Band1_Hadamard_14208_PaleyI`; `2026-10-08_Band1_Hadamard_14160_PaleyI`; `2026-10-08_Band1_Hadamard_14144_PaleyI`; `2026-10-08_Band1_Hadamard_14108_PaleyI`; `2026-10-08_Band1_Hadamard_14088_PaleyI`; `2026-10-08_Band1_Hadamard_14084_PaleyI`; `2026-10-08_Band1_Hadamard_14072_PaleyI`; `2026-10-08_Band1_Hadamard_14052_PaleyI`; `2026-10-08_Band1_Hadamard_14012_PaleyI`; `2026-10-07_Band1_Hadamard_14000_PaleyI`; `2026-10-07_Band1_Hadamard_13968_PaleyI`; `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`;

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2026-10-06_Band1_Hadamard_13368_PaleyI; 2026-10-06_Band1_Hadamard_13340_PaleyI;
2026-10-06_Band1_Hadamard_13332_PaleyI; 2026-10-06_Band1_Hadamard_13328_PaleyI;
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2026-10-03_Band1_Hadamard_12240_PaleyI; 2026-10-03_Band1_Hadamard_12228_PaleyI;
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2026-10-03_Band1_Hadamard_12164_PaleyI; 2026-10-03_Band1_Hadamard_12144_PaleyI;
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2026-09-30_Band1_Hadamard_11472_PaleyI; 2026-09-30_Band1_Hadamard_11468_PaleyI;
2026-09-30_Band1_Hadamard_11448_PaleyI; 2026-09-30_Band1_Hadamard_11444_PaleyI;
2026-09-30_Band1_Hadamard_11424_PaleyI; 2026-09-30_Band1_Hadamard_11412_PaleyI;
2026-09-30_Band1_Hadamard_11400_PaleyI; 2026-09-30_Band1_Hadamard_11384_PaleyI;
2026-09-30_Band1_Hadamard_11352_PaleyI; 2026-09-30_Band1_Hadamard_11312_PaleyI;
2026-09-30_Band1_Hadamard_11300_PaleyI; 2026-09-30_Band1_Hadamard_11288_PaleyI;
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2026-09-29_Band1_Hadamard_11244_PaleyI; 2026-09-29_Band1_Hadamard_11240_PaleyI;
2026-09-29_Band1_Hadamard_11172_PaleyI; 2026-09-29_Band1_Hadamard_11160_PaleyI;
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2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;

2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_14520_paleyI/`.
Cutoff. 2026-10-13T02:44:55Z (~10:44PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14520 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14519 (prime ===3 mod 4). 14520 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14520: Paley-I **PASS** ($q = 14519$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7259=7*17*61 NOT a prime power; ===3 mod 4 needs ===1); Kronecker **FAIL/SKIP** ($14520 = 4 \cdot 3630$; $3630 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3630 = 2 \cdot 3 \cdot 5 \cdot 11^2$; do NOT invent H_3630). **14520** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture; 14520 is a long-known Hadamard order and this is a finite on-box verification, NOT a new-order claim. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14508+14504+14480+14476+14460+14448+14432+14428+14424+14420+14412+14408+14388+143** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14520**.

- H_{14520} = Paley-I ($q = 14519$) – classical SPK
- Paley-II for 14520: **FAIL/SKIP** ($q = 7259 = 7 \cdot 17 \cdot 61$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 14520: **FAIL/SKIP** ($4 \cdot 3630$; $3630 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3630 = 2 \cdot 3 \cdot 5 \cdot 11^2$)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14520.py`

Results. overall_pass=True; units 791/791; row blocks 114/114 (block size 128); wall 5458.1356s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-13T02:44:55Z (~10:44PM ET)

Arithmetic

- Target $n = 14520 = q + 1$ with $q = 14519$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7259 = 7 \cdot 17 \cdot 61$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $14520 = 4 \cdot 3630$; $3630 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3630 = 2 \cdot 3 \cdot 5 \cdot 11^2$ — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14520 Paley-I	Paley-I q=14519	0	5419.6934
H_14520 Paley-II	FAIL/SKIP (q=7259=7*17*61 NOT prime; ===3 mod 4 needs ===1)	–	–
H_14520 Kronecker	FAIL/SKIP (4*3630; 3630===2 mod 4 NOT Hadamard-admissible; 3630=2*3*5*11ˆ2)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
14520 in SPK	**True** (Paley-I q=14519)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)
- 14520 is a long-known Hadamard order (Paley 1933 family); finite on-box verification only, NOT a new-order claim

Next optional

NEXT direct route after 14520 (FIRST any direct = Paley-I): q=14543->n=14544 Paley-I DIRECT (prime ===3 mod 4; Paley-II q2=7271=11*661 composite -> FAIL/SKIP; Kronecker 14544=4*3636; 3636===0 mod 4 order-admissible but H_3636 has NO direct classical Paley route -> FAIL/SKIP note-only; template hadamard_14520_paleyI / hadamard_14480_paleyI) | Kronecker-only NOTE 14528=4*3632 (H_3632 via Paley-I q=3631; note-only, NOT built) | then Paley-I q=14551->n=14552 | first Paley-II after 14520 = q2=7297->n=14596 | skipped Paley-II DIRECT orders below 14520 still unbuilt (template hadamard_14508_paleyII / hadamard_14476_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)